

CS-370 SPRINT 1 REVIEW

VendorShield

AI-Driven Vendor Compliance & Contract Audit Platform

SCRUM MASTER

Coralynn Yang

PRODUCT OWNER

Leah Loukedis

ENGINEERING TEAM

Chloe P., Ruilin C., Sihao Z., Danny C.



Addressing Proposal Notes

"Scope is ambitious."

Instructor concern regarding sprint containment.

→ Decomposed into 6 binary stories.

Zero partial credit. Shipped testable, isolated units.

"Map roles clearly."

Instructor requested explicit Scrum role mapping.

→ Formal Scrum structure adopted.

Coralynn (SM), Leah (PO). All 6 members act as Developers.

Foundation & Auth

COMMITTED VS. COMPLETED

39

Story Points Shipped

100% Delivery Rate



Project Scaffolding

Docker Compose (6 services) and GitHub Actions CI.



Multi-Tenant Data Schema

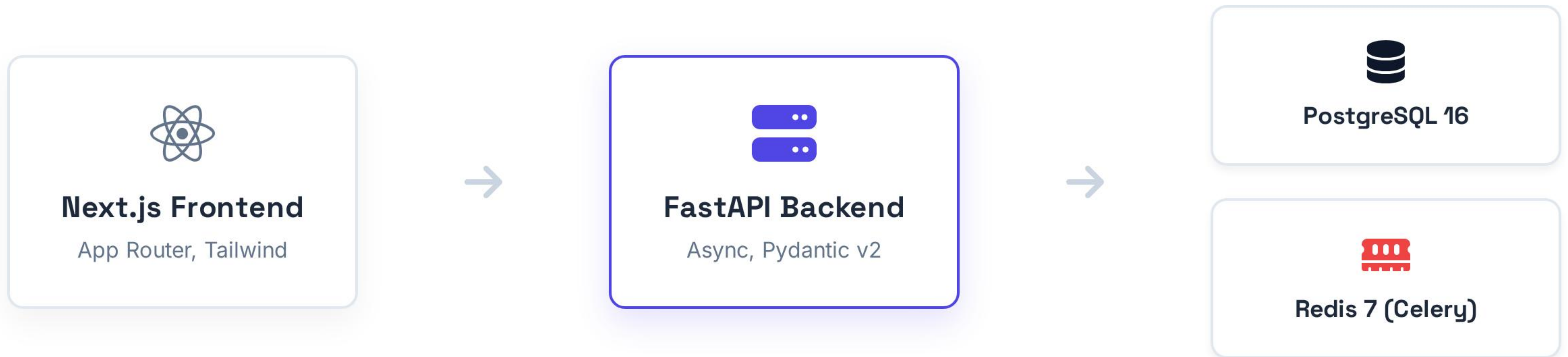
10 ORM tables, 17 enums, tenant_id isolation.



Authentication System

Dual-token JWT flow and frontend dashboard shell.

Core Infrastructure



Bottlenecks & Resolutions

BOTTLENECK	ROOT CAUSE	RESOLUTION
asyncpg Incompatibility Database	Celery workers are synchronous. asyncpg requires a running event loop.	Built SyncSessionFactory using psycopg2 exclusively for Celery tasks.
Docker Networking Infrastructure	Frontend cannot reach API at localhost inside Docker bridge network.	Configured Next.js rewrites to proxy /api/* to http://api:8000.
JWT Payload Constraints Security	Tenant isolation requires DB lookup without tenant_id in token.	Structured custom payload containing sub, tenant_id, and role.

Team Contributions

TEAM MEMBER	PRIMARY SPRINT 1 DELIVERABLES	STATUS
Coralynn Yang	Owned infrastructure layer: Docker Compose (6 services) and GitHub Actions CI.	100%
Leah Loukedis	Built frontend authentication pages and dashboard shell with CSS design tokens.	100%
Chloe Peyrebrune	Architected multi-tenant database schema, 10 SQLAlchemy models, and Alembic config.	100%
Ruilin Chen	Implemented JWT dual-token authentication flow and bcrypt password hashing.	100%
Sihao Zhang	Built tenant isolation middleware and immutable audit logging systems.	100%
Danny Chen	Engineered global API error handler, health check endpoints, and Swagger UI.	100%

METRICS

Velocity Output

4,200

LINES OF CODE

10

DATABASE TABLES

8

API ENDPOINTS

Sprint 2 Goals



Contract Ingestion

Multipart file upload with SHA-256 integrity hashing.



Text Extraction Pipeline

Celery workers parsing PDFs (pdfplumber) and DOCX.



Vendor Management UI

Full CRUD interface with grid layout and search.



Sustainability & Discussion

Sustainability

- ✓ Pinned versions in requirements.txt.
- ✓ Docker guarantees reproducible environments.
- ✓ Health endpoints support production orchestrators.

Discussion Prompt

"We use a shared database with tenant_id isolation instead of separate databases."

What approaches would you recommend for rigorously testing this isolation to prevent cross-tenant data leaks?



Thank You

We are happy to answer any questions or discuss
our architectural choices in more detail.

Image Sources



<https://www.unite.ai/wp-content/uploads/2026/08/ai-writing-generators-hero-1920x1080-1.png>

Source: www.unite.ai